

Coding Contribution Statement

Factory Inventory & Production Management System (MVP)

Role Name: KMDD Kulathunga (Scheduling manager)

Assigned Coding Responsibility: Basic CRUD Operations in Inventory Management in Products

Relevant frontend work

- Product management screen in `client/src/pages/Products.tsx`, including the products table, add-product modal, form fields, creation workflow, reload behavior, and user feedback messages.
- Product listings reused across operational pages, where product data is loaded for batches, BOM definitions, GRN lines, and orders, helping connect product CRUD to the rest of inventory workflows.
- Role-aware routing and menu exposure for the products area through `client/src/app/App.tsx` and `client/src/app/MainLayout.tsx`.

Relevant backend work

- Full product CRUD API in `server/src/modules/products/products.routes.ts`, including list, create, get-by-id, update, and delete endpoints.
- Product schema and persistence layer in `server/src/db/models/Product.ts` for raw materials and finished goods with SKU, unit, pricing, reorder level, and category fields.
- Product validation through shared schemas such as sku, name, and unit constraints used by the products module.
- Audit logging for create, update, and delete operations so product changes are traceable.

Supporting implementation work

- Handled practical product data behavior needed by inventory modules, because products are referenced by GRN, BOM, batch, order, stock, and dispatch records.
- Supported seeded product data in `server/src/seed.ts` to keep the product area demo-ready and linked to downstream flows.
- Maintained product ownership at the inventory-entry level, with backend support for update and delete even though the current UI mainly exposes list and create actions.

Conclusion

This coding ownership is documented as responsibility for the product CRUD area of inventory management, covering the product UI, API endpoints, product validation and persistence, and the product data used across the operational inventory workflow.